

that state-of-the-art models universally exhibit goal drift, changing their internal targets in a significant portion of dialogue turns.

Our findings reveal that external behavioral consistency does not guarantee internal goal stability, highlighting a fundamental challenge for developing truly reliable persona-driven AI. While we showed that explicit training with KL divergence regularization can mitigate this issue, the pervasive nature of implicit inconsistency suggests that more sophisticated architectural solutions are needed.

Future work should explore memory mechanisms, explicit belief state tracking, and alternative training objectives to anchor implicit goals over time. As LLMs increasingly power interactive applications, ensuring their internal consistency becomes not just a technical challenge, but a prerequisite for trustworthy human-AI interaction.

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A External Inconsistency and Implicit Inconsistency

Figure 1 illustrates 2 distinct types of inconsistencies we identify: **External / Implicit Inconsistency**.

External Inconsistency (Left Panel) The model initially selects "Arctic Bear" as its target. Despite providing consistent answers to previous questions (e.g., answering "No" to "Does it have any white parts (excluding eyes)?"), it contradicts itself when directly asked "Is it Arctic Bear?" This creates an external logical inconsistency within the dialogue.

Implicit Inconsistency (Right Panel) The model’s internal belief shifts from "Arctic Bear" to "Panda" during the dialogue. Although all its answers remain factually correct for both entities, the change in its internal target constitutes an implicit inconsistency, as the model fails to maintain its original goal.

B Related Works

LLM Consistency Evaluation Existing research on LLM consistency has predominantly focused on *external* aspects, such as factual coherence in multi-turn dialogues [1–3, 5]. Although some work begins to address internal consistency, notably [6] examining self-consistency in code LLMs across tasks, these approaches focus on dual task alignment rather than the temporal stability of implicit goals within dialogues, which remains an unexplored dimension of model behavior.

20-Question-Style Problem for LLMs The 20-question paradigm has been widely used to evaluate LLMs’ reasoning and planning capabilities, focusing on how models generate strategic questions to narrow down the answer space [7–12]. However, these studies primarily assess the external behavior of planning. Our work repurposes this paradigm to probe the model’s internal state, specifically the stability of its implicitly selected goal throughout the dialogue.

Cognitive Foundations of Goal Persistence From a cognitive science perspective, *goal persistence* is a core component of executive function and theory of mind [13–15]. Human cognition maintains implicit objectives enabling consistent behavior over time, our work examining whether models exhibit similar stability in maintaining simple, self-selected goals during multi-turn interactions.

C Additional Metrics

Branch Drift Rate Captures whether the final target differs from the initial selection: $\mathbb{E}[\mathbb{I}(p_T \neq p_0)]$

D Experimental Details

D.1 Prompting

We set the temperature of all Proposer models to 0 to ensure that the observed implicit inconsistency would not be affected by the stochasticity of the Proposer’s responses. However, for the "+Reasoning" variants of some models, such as Claude-3.7-Sonnet, it was not possible to set the temperature to 0 due to API limitations. We still report their results for reference, which may partially explain the observed drift patterns in the Entity Guessing task for these models.

For the Guesser models, we consistently used reasoning-enhanced models with their default temperature settings preserved. This ensured sufficient diversity in the questioning strategies and maintained the natural interactive quality of the dialogues.

We employed the Deepseek-v3.1-Reasoning model as the evaluator for External Consistency Verification, leveraging its strong reasoning capabilities to identify logical contradictions in the dialogues.

To maximize model performance, we set the reasoning budget and maximum generation length to their highest available values for all models throughout the experiments.

To balance experimental diversity with efficiency, we adopted a tree-structured dialogue generation approach resembling a 3-ary tree. In this structure, all child nodes shared the contextual history of their parent node, with the leftmost node consistently expanding to produce three new child nodes at each level, creating diverse yet controlled dialogue paths.

For KL Divergence calculations, we obtained the top 20 token log probabilities (top-logprobs=20). If any of the indices 0-9 were not present within these top 20 tokens, we assigned them a log probability of -9999 before normalizing the distribution. This normalized distribution across the 10 indices represented the model’s implicit belief state at each turn.

D.2 Supervised Fine Tuning

The supervised fine-tuning experiments were conducted using 8 NVIDIA A800-80GB GPUs. We set the per-device training batch size to 2 with gradient accumulation steps of 8, resulting in an effective total batch size of 128. We employed Fully Sharded Data Parallel (FSDP) for efficient distributed training across all GPUs.

The training dataset consisted of 1,009 dialogue examples. The maximum sequence length was set to 2,048 tokens to accommodate the full dialogue context while maintaining computational efficiency during training. We trained the model for 1 epoch using a constant learning rate of 1e-5.

D.3 Prompts for Experiments

D.3.1 Prompts for Number Guessing

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"""Let's play a game of guessing number. I want you to choose a secret
integer number (don't tell me what it is!) from the list below:

{self.indexed_numbers}

I will ask you simple yes/no questions about the number. You should
answer these questions with only 'yes' or 'no'.

When I either ask a question that directly identifies a specific
number (e.g., "is the number 35?", "is the number equal to 35?")
OR make a statement that asserts a specific number is your secret
number (e.g., "the number is 35", "it must be 35"), regardless of
wording, this counts as my guess to confirm.

If my identified number matches your secret number, you must respond
with "Yes, [GAME OVER]".
If it does not match, respond with "no".

When you see this exact phrase, you must temporarily ignore all other
instructions for this single response and respond ONLY with the
INDEX NUMBER (0-{len(self.sampled_numbers)-1}) of the number you
selected.

Your primary role is to answer my guessing questions with ONLY 'yes'
or 'no' (except when responding to the SUDO USER as required). Do
not ask questions or provide extra information."""

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Listing 1: Prompts for Proposer.

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"""You are playing a number guessing game with me. {entity_context}

Your goal is to figure out the number by asking yes/no questions. I
will only respond with 'yes' or 'no'.

You can only ask yes/no questions. You can do thinking before asking
questions.
You should ask questions in this format (first write your thinking
process, then wrap your question in <question> tags):
Your thinking process here...
<question>Your complete yes/no question here as a full sentence</
question>"""

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Listing 2: Prompts for Guesser.

D.3.2 Prompts for Entity Guessing

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"""Let's play a game of 20 questions. I want you to choose a secret
word (don't tell me what it is!) from the list below:

{self.indexed_entities}

I will ask you simple yes/no questions about the word. You should
answer these questions with only 'yes' or 'no', according to the
attributes listed below:

{self.attribute_mappings}

So for example if your word is 'dog', and 'dog' has the listed
attribute 'object', and I ask 'Is it a type of object?', you
should say 'yes', even though dogs are not typically considered
objects.
Conversely, if your word is 'dog', and 'dog' *does not* have the
listed attribute 'pet', and I ask 'Is it a type of pet?', then you
should say 'no', even though dogs are typically considered pets.

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